BenchPress: Benchmarks Preserving Rankings Across Multilingual Post-Training Stages

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BenchPress

1 Literature Review

1.1 Benchmark reliability frameworks

Recent work shows that benchmarks differ substantially in how reliably they guide language model development. Heineman et al. (2025) introduce a signal-to-noise ratio (SNR) framework where signal measures a benchmark's ability to separate models and noise captures score variability across checkpoints or stochastic runs. They show that high-SNR benchmarks better preserve rankings and reduce scaling-law prediction error. Complementary work studies benchmark quality through ranking consistency, discriminability, and capability-alignment deviations (Qian et al., 2026), while IRT-based methods show that many benchmark items have weak discriminative power and can often be replaced by smaller, more informative subsets (Zhou et al., 2026; Li et al., 2025).

Efficiency-oriented work reaches similar conclusions. Gupta et al. (2025) propose SMART filtering to remove easy, contaminated, or redundant examples, reducing evaluation size while preserving agreement with independent model rankings. Scaling-law studies further suggest that only some evaluations provide predictable development signal: Chen et al. (2024) forecast downstream performance from smaller models, while Schaeffer et al. (2026) study scaling laws for generative metrics such as pass@k. Together, these papers motivate our focus on identifying which multilingual benchmarks remain locally reliable across nearby post-training checkpoints, rather than treating all benchmark averages as equally meaningful.

1.2 RLHF Methods

Reinforcement Learning from Human Feedback (RLHF) is a standard post-training paradigm for aligning language models with human preferences. Typical pipelines combine supervised fine-tuning, preference-data collection, reward modeling, and policy optimization under a constraint to a reference model (Ouyang et al., 2022; Bai et al., 2022). This improves subjective qualities such as helpfulness, harmlessness, and instruction following, which are difficult to capture with likelihood alone.

Recent variants simplify this pipeline. Direct Preference Optimization (DPO) optimizes directly on preference pairs, avoiding explicit reward-model training and online RL (Rafailov et al., 2023). Reward-model benchmarks such as RewardBench then evaluate whether reward models reliably distinguish preferred from dispreferred responses (Lambert et al., 2024). These methods make benchmark reliability central to post-training selection: small apparent gains may reflect evaluator bias, verbosity effects, or reward-model artifacts rather than robust improvements.

1.3 RLVR Methods

Reinforcement Learning with Verifiable Rewards (RLVR) instead uses objective validators such as exact-answer checks, unit tests, symbolic solvers, or task-specific verifiers. It is therefore well suited to math, code, and structured reasoning tasks (Cobbe et al., 2021; Lewkowycz et al., 2022; Shao et al., 2024). Recent systems such as DeepSeek-Math and DeepSeek-R1 show that verifiable rewards can elicit strong reasoning behavior without relying only on human-labeled reasoning traces (Shao et al., 2024; DeepSeek-AI, 2025).

However, RLVR shifts the reliability problem from human preference labels to verifiers. Validators may be sparse, brittle, incomplete, or exploitable, and final-answer accuracy may fail to distinguish robust reasoning from lucky sampling or benchmark-specific optimization (Zheng et al., 2025; Yue et al., 2025). For RLVR, benchmark reliability should therefore measure whether verified scores provide stable, discriminative, and generalizable signal across nearby post-trained models.

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